BENNETT UNIVERSITY
THE TIMES GROUP

Enrolment No: ______________________ Name of Student: ______________________

Department/ School: ____

END-TERM EXAMINATION, ODD SEMESTER DECEMBER 2023

COURSE CODE	CSET-301	MAX. DURATION	2.5 HRS
COURSE NAME	Artificial Intelligence & Machine Learning		
PROGRAM	BTech 5 Semester	TOTAL MARKS	35

Mapping of Questions to Course and Program Outcomes										
Q.No.	A1	A2	A3	A4	A5	A6	A7	A8	A9	A10
CO	1	2	3	1	2	3	3	3		
PO	1-2	1-2	1-2	1-2	4-5	4-5	4-5	4-5		
BTL	1	2	3	1	2	3	3	4		

GENERAL INSTRUCTIONS: -

1. Do not write anything on the question paper except **name, enrolment number** and **department/school.**
2. Carrying mobile phones, smartwatches and any other non-permissible materials in the examination hall is an act of **UFM.**
3. **Simple calculator is allowed.**

COURSE INSTRUCTIONS:

a) Solve any seven questions.
b) All questions carry equal marks.

SECTION- A

Max Marks: 35

A1. Describe Linear Regression and explain different performance metrics with formula used for evaluating its performance. **(Marks: 5)**

A2. Explain Bayes' Theorem and its role in Naive Bayes. Elaborate the key assumptions underlying the Naive Bayes algorithm? **(Marks: 5)**

A3. Support Vector Machine (SVM) exhibits promising performance for linearly separable data. However, in real-life most of the data are not linearly separable. Explain SVM and how it can help to solve the above problem. **(Marks: 5)**

A4. Describe overfitting and explain how it affect a model's ability to generalize to unseen data? Contrast L1 and L2 regularization methods used to reduce overfitting. **(Marks: 5)**

A5. Using the following confusion matrix, calculate accuracy, F1-score, precision, recall, and AUC.

(Marks: 5)

n=165	**Predicted NO**	**Predicted YES**
Actual NO	50	10
Actual YES	5	100

A6. In the midst of year 2020 following data is collected from the symptomatic COVID patients. After the deep analysis of collected data, doctors asked the research scientist to create a decision tree for early identification of suspected COVID patients. Write detailed steps and create a decision tree (use information theory) to help the doctors combat COVID-19 (up to first branching). **(Marks: 5)**

Patient ID	Fever	Sore Throat	Shortness of Breath	Class
P1	High	Yes	Yes	COVID-19
P2	High	Yes	No	Quarantine
P3	High	No	Yes	Quarantine
P4	Medium	Yes	Yes	COVID-19
P5	Medium	Yes	No	Quarantine
P6	Medium	No	Yes	Quarantine
P7	Low	Yes	Yes	COVID-19
P8	Low	Yes	No	Quarantine
P9	Low	No	Yes	Quarantine

A7. The following table shows the obtained marks and grade by different BTech students in different CTs of AIML subject in July to December 2023. Unfortunately, Aman got a major accident during the End-Term exam. The subject teacher decided to assign a grade to Aman based on his class-test marks using K-nearest neighbour (K=3) classifier. Write steps to find the Aman's grade. **(Marks: 5)**

Table: Class test marks and grade of the other students.

Student's Name	CT-1 Marks	CT-2 Marks	Final Grade
Riyaan	14	12	A
Sushant	13	12	A
Monika	10	12	B
Yuvi	14	5	B
Kamal	8	7	C
Mohit	9	6	C
Aman	**12**	**9**	**??**

A8. Cluster the following eight points (with (x, y) representing locations) into three clusters: A1(2, 10), A2(2, 5), A3(8, 4), A4(5, 8), A5(7, 5), A6(6, 4), A7(1, 2), A8(4, 9), Initial cluster centers are: A1(2, 10), A4(5, 8) and A7(1, 2). The distance function between two points a = (x1, y1) and b = (x2, y2) is defined as P (a, b) = |x2 – x1| + |y2 – y1|. Use K-Means Algorithm to find the three cluster centres after the Second iteration. **(Marks: 5)**